

SensoryScape: Context-Aware Empathic VR Photography

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ABSTRACT

SensoryScape presents a novel approach to context-aware empathic interactions (CAEIs) using a VR head-mounted display (HMD) integrated with various biosensors. This innovative system addresses the need for personal, emotionally resonant, and empathic digital experiences by dynamically responding to users' real-time emotional and situational context. SensoryScape harnesses the Galea VR HMD, a device with integrated physiological sensors that track biosignals such as brain activity, skin responses, muscle movements, and heart rate variability. These real-time biosignals, along with situational context, help estimate the user's emotional state within its context. The VR environment and empathic companion adapt to these cues, ensuring an optimized experience for the user's current emotional state. The demonstration of SensoryScape highlights the potential of VR HMDs with integrated physiological sensors for crafting empathic digital experiences. Our research contributes to immersive, adaptive, and user-focused empathic interactions, marking a promising shift in human-computer interaction.

KEYWORDS

Virtual Reality, Biosignals, Empathic Computing, Context-Aware

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1 INTRODUCTION

The SensoryScape demonstrates how using a VR HMD integrated with various biosensors can create context-aware empathic interactions (CAEIs). CAEIs present a transformative approach to human-computer interaction, providing an avenue for more personal and emotionally resonant digital experiences. These interactions aim to bridge digital landscapes 'empathy gap' by dynamically adjusting to the user's emotional and mental states in real-time. With potential applications spanning sectors such as healthcare, education, and customer service, CAEIs stand to revolutionise digital interfaces by providing a deeper understanding of individual needs and enhancing user experiences.

Detecting the user's emotional state in real time is crucial to CAEIs. This has been made possible by integrating biosensing technologies into Virtual Reality Head-Mounted Displays (VR HMDs). The HP Omnicept [HP 2023] captures bio-indicators such as eye gaze and PPG, while the PhysioHMD [Bernal et al. 2018] incorporates sensors like EMG, EEG, and EDA into VR HMDs. Prior studies have validated these sensors for predicting emotional states in VR [Gupta et al. 2020].



Figure 1: Galea HMD (bottom view): Red shows EEG electrodes, and Green shows EMG, EDA, and PPG sensors

Understanding a user's physiological and emotional state is just the beginning. Our research delves deeper into situational contexts, like landmark reach time or photo availability, enriching our grasp on emotional context. We integrate this context with physiological data using the self-projected Appraisal technique [Wispé 1986].



Figure 2: SensoryScape VR view. The user’s emotional state changes the environment colour tint and also the colour and speaking tone of the circular empathic companion

Utilizing the Galea VR HMD [Bernal et al. 2022] (Fig. 1), which offers an extensive sensor range including EEG, EOG, and HRV, we’ve crafted a CAELx platform in VR.

Our showcased VR Photo Walk features an emotion-responsive environment and an empathic companion, both adapting to real-time user emotions. The strength of our research combines Galea’s sensory data, contextual insights, and a dynamic SensoryScape application, presenting a richer digital interaction that encourages a state of flow.

2 DESIGN AND USER EXPERIENCE

In this demo, participants embark on a monument hunt within a virtual island environment, tasked with capturing the optimal virtual photographs of two distinct monuments. Constraints bound their photographic efforts: they can take only eight shots and must do so within a limited time frame. Assisting them in this endeavour is a virtual companion, facilitating navigation, timing, and photographic choices. The underlying objective is to probe the effects of context-aware empathic interactions, encompassing emotion-adaptive environments and virtual companions equipped with empathic dialogues, on inducing and maintaining the flow state of the user.

Beginning at an island campsite, participants have 5 minutes to find and photograph two monuments using a VR camera, activated by raising their faces and pressing a VR controller button. After each shot, a virtual companion updates on the shots left and asks whether to keep the current photo. Meanwhile, the Galea HMD streams physiological metrics like EEG, HRV, and EDA to an emotion recogniser. The system then adjusts VR visuals and the companion’s tone based on the detected emotion. Participants can ask the companion for directions to the next monument, receiving empathic guidance. The companion also issues time alerts, notably at the halfway point, 60 seconds, and 10 seconds left, elevating stress. Upon detecting significant stress, the companion offers soothing, empathic dialogues. This real-time adaptation enhances the user experience, maintaining their immersion and flow state.

Concluding the SensoryScape demo, participants can select and cherish their most favoured monument photograph. This is then transmuted into a tangible postcard, serving as a memento of their

virtual expedition and reinforcing the blend of virtual immersion with tangible memory.

3 IMPLEMENTATION

The demo comprises three components: the BioSignal Processor (BP), the Emotion Predictor (EP), and the Empathic Adaptor (EA). BP uses the Galea headset to gather EEG, EMG, EDA, and PPG, which are streamed to a Python signal processor via the Lab Streaming Layer (LSL) protocol. After data cleaning and preprocessing, features from these bio-signals are forwarded to the EP’s machine-learning emotion model. EP identifies emotional states: ‘excited’, ‘stressed’, ‘bored’, and ‘serene’ or ‘relaxed’. The LSL protocol conveys these emotions to EA in the VR.

Besides interpreting emotions, EA assesses context—like completion time and photo count—to select empathic dialogues. Users’ emotions manifest visually as tint filters on their view: yellow for ‘excited’, red for ‘stressed’, blue for ‘bored’, and green for ‘relaxed’, based on Itten’s Color system and Russell’s affect model [Ståhl et al. 2005]. Post-pilot feedback led to specific tones of each emotion using Microsoft’s Speech SDK. The demo was built on Unity3D version 2020.3.28f and showcased through Galea VR HMD on SteamVR.

4 RESEARCH PLANS

SensoryScape demonstrates the use of physiological sensing and empathic feedback in a photo-taking VR application; the most important aspect is developing a sensing and ML platform that could be applied to many different VR application types. In the future, we will collect user physiological responses when interacting in VR to improve our emotion-recognition system. The platform and data will be publicly available to the research community to help create the next generation of emotionally responsive VR with CAELx. In the future, we look forward to further exploring the application space. We are particularly interested in understanding the impact of empathic interactions on user emotions, engagement, immersive, flow state, and cognitive load in VR.

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